

data-model

Phase 1 — Data Model

Feature: Session record + QA readiness · **Date:** 2026-09-08

Entities, invariants, and states. The wire form is in [contracts/](#).

Four invariants govern everything, and each exists because its absence was a measured defect:

- **I1 — One derivation, many consumers.** A fact is computed once. Two implementations that agree today produced this platform's 819-listed-vs-817- openable catalogue bug.
- **I2 — Absence is a state, not a null.** Every missing case is named and reported with a reason. A synthesised value standing in for an absent one is how 814 mined areas acquired names they never had.
- **I3 — A snapshot licenses only itself.** A cached copy may assert what it holds, never a determined negative about the source it can no longer see.
- **I4 — A proposal is not a decision.** A machine-drafted verdict may not move a published figure.

SessionRecord

One chapter's record, cumulative over that session and every prior one.

Field	Notes
chapter	The chapter this record belongs to.
prior_chapters	The chapters folded in. Cumulativeness is data, not an assumption.

Field	Notes
sections	The four, keyed by name.
section_order	meeting_notes, open_questions, todo, next_meeting — one order, so no surface invents its own.
plan	What this session intended to cover.
carry_forward	Points moving to the successor, or the outbox.
privacy	The boundary statement carried with the record itself.

Invariants

- All four sections present (FR-001). Cumulative over prior sessions (FR-002).
- **Both the embedded chapter key and the standalone routes read the same derivation (I1).** A gate asserts they agree; agreeing today is what drifts.

Section

Field	Notes
own_state	authored OR empty.
own_count	Items originating in this session.
carried_count	Items arriving from earlier sessions.
items[]	Both, together.
reason	Required when own_state is empty.

Invariant (FR-004, I2): a section with nothing to report **says so with its reason. An empty section and an unwritten section must never be indistinguishable** — this is the single property that lets a QA team tell a defect from a blank.

The own_count/carried_count split is load-bearing. Two sections currently carry own_count=0 with an authored reason. Without the split they would read as productive sessions.

RecordItem

Field	Notes
id, text	
origin_chapter	Which session raised it (FR-003).
slip_count	How many times it has been carried (FR-010).
marker	CARRIED - FORWARD when carried (FR-009).
marker_label	e.g. CARRIED - FORWARD x2 — the count is visible, not merely stored.
resolution	Open questions only: answered_in_chapter / answered_by_research / unanswered. The client already types this; the wire must not narrow it.

SessionPlan

Field	Notes
basis	predecessor_next_points or undetermined.
coverage_state	undecided until a human confirms (I4).
coverage_source, coverage_reason	Where a verdict came from, or why none exists.
planned_count, covered_count, carried_count, dropped_count	

Invariants

- Every planned point resolves to **covered** / **carried** / **dropped-with-reason** (FR-007). **vanished is 0 by construction** — Resolve has no fourth branch.
- **A drop without a reason is refused and carries instead**, so “dropped” cannot become a quiet disposal route.

- **basis: undetermined computes no coverage percentage.** Chapter 01 has no predecessor and no recorded intent; a percentage over an invented plan would be a fiction with a denominator.

CoverageProposal (*new*)

Field	Notes
point_id	
verdict	covered / not_covered / cannot_determine — a first-class answer.
evidence[]	Passage and section ids the verdict rests on.
confidence	The proposer's own.
status	Always PROPOSED until a human confirms.

Invariant (I4, FR-007a): a proposal **MUST NOT** move `coverage_state` or any served count. A check asserts the served state is still undecided after a proposal exists — that assertion is what makes the rule mechanical rather than aspirational.

CarryForward

Field	Notes
to_chapter	Successor, or null when this is the outbox.
is_outbox	True for the latest chapter, which has no successor (FR-011).
reason	Why these points are here.
max_slip_count	The worst slip in the set — a point slipping repeatedly is surfaced, not buried.
points[]	

Lesson

Field	Notes
id, ord, title	ord drives one server-side sort; prev/next use the same slice.
body	The teaching content. Declared upstream in the kit , optional there by deliberate choice — a materials-only lesson is legitimate, so “every lesson has a body” is a <i>consumer</i> policy.
summary	A one-line blurb. Kept distinct from body ; collapsing them was the semantic lie that caused 339 bodies to be discarded.
estimated_minutes, materials[], required_for_assessment	

Invariant (FR-013): every served lesson of a published area carries non-empty content. Asserted by a gate with **two arms** — in-process and live — because in-process alone is a source measurement.

Assessment · Question · Attempt

Unchanged from spec 004 except three properties this feature added:

- **The taking projection has no answer field at all** — built as a map, not as the domain type with fields omitted. Omission depends on every future field remembering to omit itself; absence does not.
- **Graded rows are withheld from the open practice route**, and the withholding **says so**. The two decks are the same authored rows: 224 of 224 match by prompt, 0 of 224 by id.
- **A malformed submission is refused**, naming the offending key, before the store is touched. responses: [] still grades — a blank paper is a real submission.

Fail-closed (I3): a nil, unreadable **or stale** catalog withholds every answer key rather than risking one.

Catalog snapshot

State	Meaning
Fresh	Source unchanged since load.
Stale	Source changed — determined negatives are refused; 503 names the changed files.
StalenessUnknown	Cannot tell. Never a pass.
NoSource	No directory to have drifted from. Negatives still stand — an in-memory test catalog is legitimately authoritative about itself.

NoSource is a separate state on purpose. Folding it into Fresh would license false negatives; folding it into Unknown would break every in-memory test.

QAScenario · EvidenceArtefact

QAScenario	
action	What the tester does.
expected	Stated before the action (FR-018).
report_procedure	How to report a discrepancy and what to capture.

EvidenceArtefact	
claim	
population	source / in_process / served — required (FR-026). The same figure over different populations is a different fact.
build_id	What state it was produced from.
corpus_fingerprint	Proof the corpus did not move mid-measurement (FR-027).
verdict	

EvidenceArtefact	
	satisfied / finding / could-not-determine — never recorded as satisfied (FR-024).